

Chan Ding Hao

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Work Experience

ByteDance - Payment Risk Data Analytics Intern

Jan 2026 - Mar 2026

- Analyzed 1M+ daily new user registration records to surface behavioral fraud signals (IP geolocation, device fingerprinting, velocity patterns), establishing baseline fraud rates and proposing feature candidates that informed a downstream risk scoring framework.
- Engineered HiveSQL queries processing 1M+ daily user registration records, applying quantile analysis across device and IP distributions to identify anomalous patterns that contributed to a downstream fraud risk scoring framework.

Razer - Data Analyst Intern (Analytics Engineering)

Jul 2025 - Dec 2025

- Built Apache Superset dashboards tracking key recommendation engine metrics (Daily CTR, Daily Revenue) using SQL, reducing stakeholder reporting time by 25% and enabling self-serve analytics for business teams.
- Built transformation models (SQL) within ELT pipelines using dbt DAGs to deliver business intelligence solutions for internal stakeholders served through Apache Superset, increasing dashboard load speed by 50%.
- Developed deduplication logic for user software data through awswrangler, SQL and pandas in AWS SageMaker AI notebooks, identifying data quality issues and providing implementation specifications to data engineering.

ONE (Ocean Network Express) - Data Analyst Intern

Jun 2024 - Dec 2024

- Engineered a CI/CD route optimization pipeline on GCP (Vertex AI, BigQuery) using Python and SQL, improving data processing speed by 10% and enabling real-time logistics decision-making across truck fleet operations.
- Identified critical data constraints (e.g., Hazmat routing rules) through stakeholder validation, incorporating findings into the optimization model to reduce total route distance by 5% and save \$500K+ USD annually.

Projects

Crypto Market Data Ingestion (In Progress)

Jan 2026 - Apr 2026 (Expected)

- Built end-to-end crypto market intelligence pipeline on AWS (EventBridge to Lambda to RDS PostgreSQL) ingesting daily BTC/USD OHLC snapshots from Kraken REST API, enabling exchange-focused business metrics analysis.
- Designed and deployed Grafana dashboards tracking 5 key exchange performance metrics — trading volume, VWAP vs. close, volume-volatility correlation, 30-day volatility regime, and drawdown — to surface actionable revenue and liquidity insights.

SPY Analytics and Forecasting - [Github](#) - [Medium](#) (1) , (2) , (3)

Aug 2024 - Dec 2024

- Wrangled and analyzed 30+ years of SPY historical and inflation-adjusted return data alongside Federal Reserve datasets to quantify long-term equity performance advantages, published as a 3-part Medium series.
- Constructed a weighted ensemble model (LSTM -PyTorch, Prophet, and Monte Carlo simulations) for time series forecasting. Predicted the log returns of SPY in the next 20 years with confidence intervals.

VCT eSports Manager Hackathon - [Github](#) - [Medium](#)

Aug 2024 - Oct 2024

- Built a Retrieval-Augmented Generation (RAG) app, utilising Bedrock, S3 and Lambda for eSports team-building, handling data scraping, preprocessing, and model ingestion.
- Engineered custom chunking strategies (with Cohere v3 Embeddings), for structured (csv) and unstructured data (md), and vector retrieval pipelines (Pinecone) enhancing LLM based player selection.
- Deployed the model on Streamlit Cloud, streamlining user interactions for seamless decision-making in eSports.

Education

Singapore Management University - Bachelor of Science in Economics

Aug 2022 - Dec 2026

Relevant Coursework: Big Data Architecture, Machine Learning & Applications, Algorithms & Programming, Visual Analytics for Business Intelligence , Business Data Management, Computational Thinking & Programming, Data Analytics with R

Skills and Certifications

Technical Skills: Python, PySpark, SQL (HiveQL, SparkSQL, MySQL, PostgreSQL), R, Bash/Linux, Git, Docker

Data Engineering & Orchestration: Airflow, dbt, Spark, Kafka

Libraries & Frameworks: NumPy, Pandas, PyTorch, scikit-learn, XGBoost, Plotly, Boto3, AWS Wrangler

Cloud & Platforms: GCP (BigQuery, Vertex AI), AWS (S3, Redshift, RDS, Lambda, SageMaker, Bedrock), Pinecone

BI & Visualization: Superset, Grafana, Tableau

Certifications & Training (Hyper-linked): [Google Data Analytics Professional Certificate](#), [HackerRank SQL \(Advanced\)](#)